

Chapter 08 Homework Answer Sheet

Name: _____ Student ID: _____ Instructor: _____ Score: _____

Part A multiple choice (15 points)

1-5 CACBA

6-10 CCBBE

11-15 CAAAC

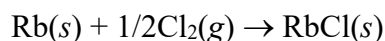
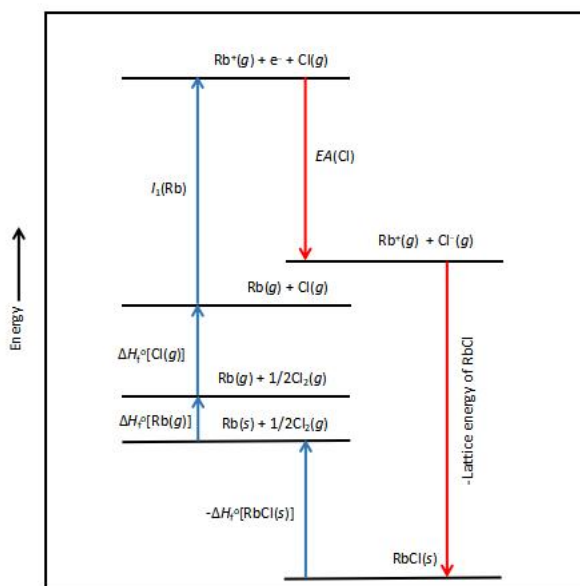
Part B short question (15 points)

16. (5 points)

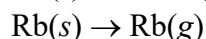
Answer:

(a) (1 point) $\text{RbCl}(s) \rightarrow \text{Rb}^+(g) + \text{Cl}^-(g)$

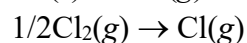
(b) (2 points)



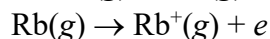
$$\Delta H_f^\circ[\text{RbCl}(s)] = -430.5 \text{ kJ/mol}$$



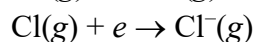
$$\Delta H_f^\circ[\text{Rb}(g)] = 85.8 \text{ kJ/mol}$$



$$\Delta H_f^\circ[\text{Cl}(g)] = 121.7 \text{ kJ/mol}$$



$$I_1[\text{Rb}(g)] = 403 \text{ kJ/mol}$$



$$EA[\text{Cl}(g)] = -349 \text{ kJ/mol}$$

Use Hess's law to calculate the lattice energy of RbCl

$$\Delta H_{\text{lattice energy}}[\text{RbCl}(s)] = -\Delta H_f^\circ[\text{RbCl}(s)] + \Delta H_f^\circ[\text{Rb}(g)] + \Delta H_f^\circ[\text{Cl}(g)] + I_1[\text{Rb}(g)] + EA[\text{Cl}(g)]$$

$$\Delta H_{\text{lattice energy}}[\text{RbCl}(s)] = 430.5 + 85.8 + 121.7 + 403 + (-349) = \mathbf{692 \text{ kJ/mol}}$$

(c) (1 point) Lattice energy is usually **endothermic**.

(d) (1 point) RbCl has **smaller** lattice energies compared to salts like SrO.

17. (2 points)

$$\Delta H_{\text{rxn}} = 2 \times 3D(\text{N-Cl}) - D(\text{N}\equiv\text{N}) - 3D(\text{Cl-Cl})$$

$$= 2 \times 3 \times 200 - (941 + 3 \times 242)$$

$$= 1.200 \times 10^3 - 0.941 \times 10^3 - 0.726 \times 10^3$$

$$= -0.467 \times 10^3 \approx \mathbf{-4.7 \times 10^2 \text{ kJ}}$$

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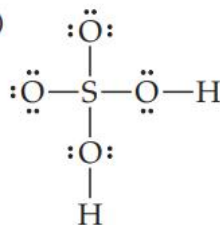
18. (6 points)

(a) (i) $[\text{Ar}]4s^23d^6$, (ii) $[\text{Kr}]5s^24d^{10}$, (iii) $[\text{Ar}]4s^23d^{10}4p^5$

(b) (i) Fe^{3+} , Fe^{2+} , (ii) Cd^{2+} , (iii) Br^-

19. (2 points)

(a)



(b)

